Roll No. | | | | | | | |

Paper Code: MSE 102

M. Tech (Structural Engineering)
End Semester Examination 2019
1st Semester

FEM in Structural Engineering

Time: Three Hours

MM: 100

Note:
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.

(2X10=20 Marks)

Q1. a Write the stiffness matrix for the vertical bar shown in fig. 1. Element area= 1.5 in^2, E=30,000,000lbs/in^2 ,Element length= 5ft. Also solve for the stress, joint displacement, and reaction force on the element

Fig. 1

b. Determine the reactions and support reactions for the uniform bar shown in fig. 2 . Given P= 300kN

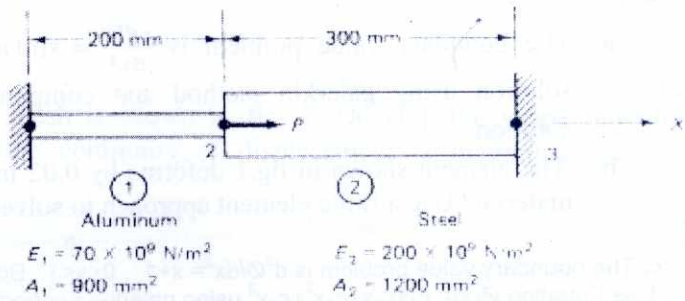
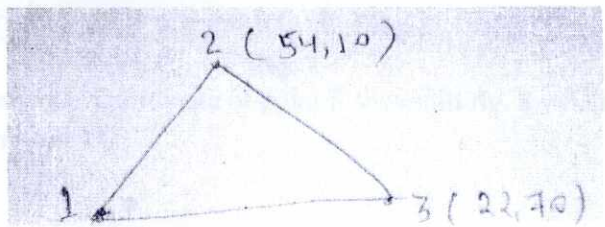


Fig. 2

c. Determine the shape function of a triangular element Q (20, 30) of a constant strain triangle (CST) element shown in fig.3 with co-ordinates 1(0,0), 2(54,10) & 3(22,70). All dimensions are in mm. If there is a point inside the CST P(X,Y) determine the co-ordinates of the point P.



Q2. (2X10=20 Marks)

(a). Compute the displacements at q_3, q_5, q_6 of a truss member shown in fig. 4. Use stiffness matrix

$$[k] = \frac{EA}{L}\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta & -\cos^2\theta & -\cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta & -\cos\theta\sin\theta & -\sin^2\theta \\ -\cos^2\theta & -\cos\theta\sin\theta & \cos^2\theta & \cos\theta\sin\theta \\ -\cos\theta\sin\theta & -\sin^2\theta & \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$$

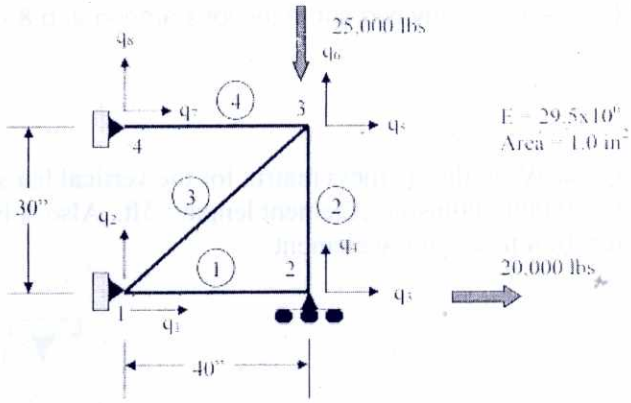


Fig. 4

b. A cantilever beam of 1m length carries a single point load at the end of the beam of 10kN. Calculate the deflection at the end of the beam by using any FEM if E=70Gpa A=500mm^2. I=2500mm^4.

c. Using Rayleigh Ritz method, calculate the deflection at centre of fixed beam of length 'L' subjected to UDL of w/unit length. Use equation of curve $y=a_1+a_2x+a_3x^2$

Q3. (2X10=20 Marks)

a. The boundary value problem is $\frac{-d^2u}{dx^2} = \sin(\pi x)$, $0<x<1$. Calculate the approximate solution using galerkin method and compare between the exact and approximate solution.

b. The element shown in fig.1 deforms by 0.02 inches in length. What is the stress in the material? Use a finite element approach to solve the problem.

c. The boundary value problem is $d^2Ø/dx^2= x+1$, $0<x<1$. Boundary condition $Ø_{x=0}=0$, $Ø_{x=1}=1$. Use Equation $Ø(x)= c_1+c_2x+c_3x^2+c_4x^3$ using galerkin method. Calculate the approximate solution.

Q4. (2X10=20 Marks)

a. The quadrilateral element shown in fig. 5 is 20mm thick and is subjected to surface force, T_x=10N/mm^2 and T_y =15N/mm^2 Determine expression for its equivalent nodal force.

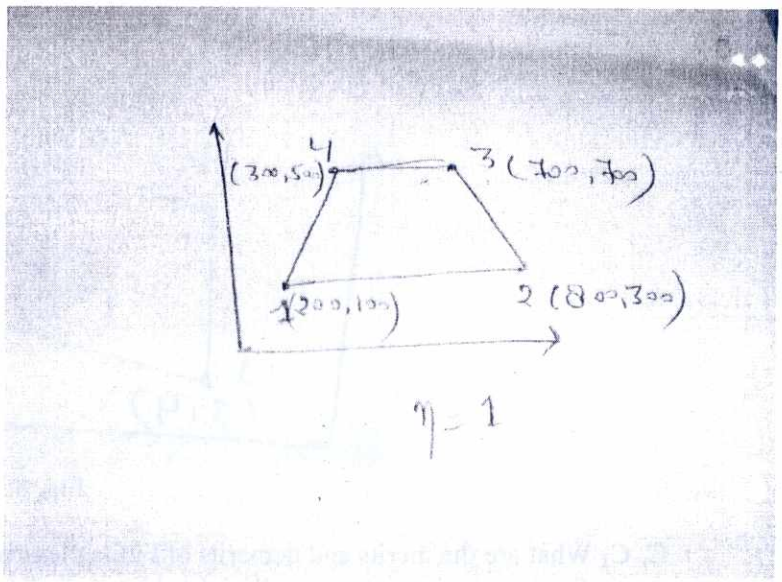


Fig. 5

b. Calculate the displacements for node 1 and 2 i.e u_1 and u_2 for fig. 6. Also calculate the reaction force at fixed support

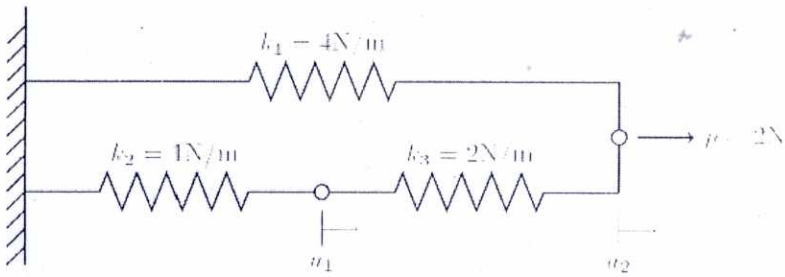


Fig. 6

c. Explain the significance of weighted residual method, Collocation method and Galerkin method.

Q5 (2X10=20 Marks)

a. A finite element mesh is shown in fig. 7. Develop the shape function for the transition element so that the continuity of displacement is maintained across the inter-element boundary.

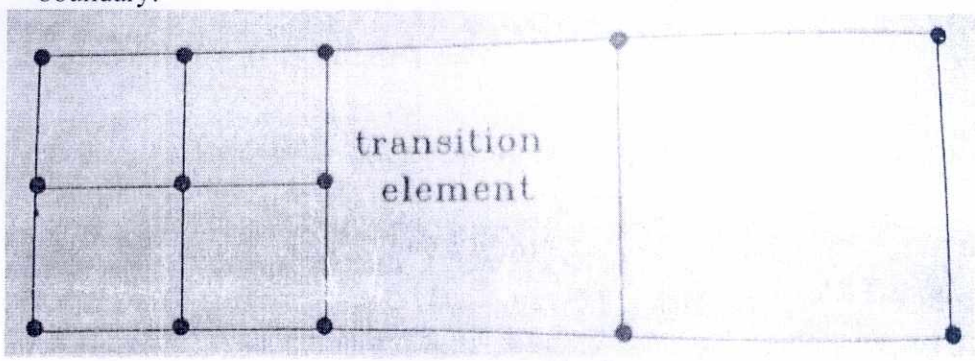


Fig. 7

b. Determine the Cartesian coordinate of point P shown in fig. 8 which has local co-ordinate ϵ=0.9 and η= 0.6

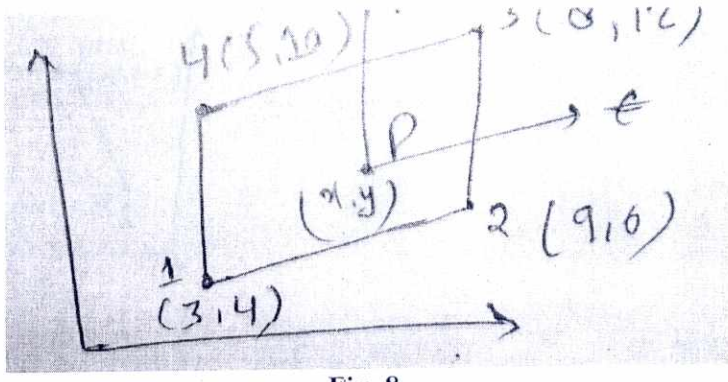


Fig. 8

C. C_1 What are the merits and demerits of Finite element method?
C_2 Explain the importance of Weighted integral statements?

Roll No. | | | | | | | |

Paper Code: MGI 102

M.Tech (Geo-Informatics)

End Semester Examination 2019

I Semester

Fundamentals of Geographic Information Systems

Time: Three Hours

MM: 100

Section A

Note: All Questions are Compulsory in Section 'A'

(20 Marks)

Q1.

a) In land use / land cover classification of Satellite imagery the error matrix resulting from randomly sampled test pixels are given below. Find out the Producer's accuracy, User's accuracy, Overall accuracy & KHAT or KAPPA statistics and also discuss the result (7)

	W	S	F	U	C	H
W	256	3	0	1	3	0
S	0	126	2	0	0	1
F	5	0	56	1	1	0
U	0	0	2	101	0	2
C	1	0	0	0	87	1
H	0	0	1	0	1	11

b) "Information plays a vital role to tourists in planning their travel from one place to other". Draw a flow chart for a Tourist GIS which works for Uttarakhand State? (7)

c) "GIS finds its application in all those area where professionals are involved" justify the statement by giving suitable application of GIS in different field? (6)

Section B

Note: Attempt all Questions from this Section, any two part from each.

(10 X 2 = 10 Marks)

Q2.

a) The data for GIS is collected from different sources explain in brief all the data sources for GIS? (10)

b) Draw the flowchart for the study of accident prone area in a city? (10)

c) Explain the History of GIS in the world and its development in India? (10)

(10 X 2 = 10 Marks)

Q3.

a) GIS have different approach of performing map overlay list them out and explain any one in detail? (10)

b) Explain the conceptual model of spatial information with the neat diagram? (10)

c) Explain in brief the Relational Model? Gives its advantages and disadvantages also? (10)

(10 X 2 = 10 Marks)

Q4.

a) What are the advantages and disadvantages of raster data model and a vector data model? (10)

b) Write a short note on: (10)

1) Map Projection and Transformation
2) Rectification and Registration

c) Draw the flowchart for site suitability analysis for urbanization? (10)

(10 X 2 = 10 Marks)

Q5.

a) Explain all the method of Spatial Interpolation? (10)

b) Draw the flowchart for the study of irrigation performance in a state? (10)

c) Write a short note on Problem Identification and Project Management in GIS Application? (10)

Roll No. | | | | | | | |

Paper Code: **VDM-102**

M. Tech (VLSI Design and System)

End Semester Examination 2019

….. Ist *Semester*

Paper Name ……Digital System Using VHDL/ Verilog ……

Time: Three Hours

MM: 100

Note:

(i) This question paper contains two sections.

(ii) Both sections are compulsory.

Section – A

Note: All questions are compulsory in section A.

Q1.

a) Write a verilog code for one bit comparator along with circuit diagram and truth table.CO1 (7 marks) (20 marks)

b) Write a verilog code for full substractor along with circuit diagram and truth table. CO1 (7 Marks)

c) What do you understand by blocking and non blocking assignment. CO2 (6 marks)

Section – B

Note: Attempt all questions from this section, opting any two parts from each.

Q2.

(10X 2 = 20Marks)

a. Write a verilog code encoder and decoder with circuit diagram and truth table. CO3 (10 Marks)

b. Write a verilog code for SR flipflop and T flipflop along with truth table. CO3 (10 Marks)

c. Write verilog code for JK flipflop and D flipflop along with truth table. CO3 (10)

Q3.

(10X 2 = 20Marks)

a. Write a verilog code for 3 bit up counter and floating point divider. CO5 (10 Marks)

b. Explain various type of operator used in verilog along with examples. CO4 (10 Marks)

c. Write a verilog code excess to BCD converter along with circuit diagram and truth table. CO4 (10 Marks)

Q4.

(10X 2 = 20 Marks).

a. Draw and explain design flow of VLSI Chip design. CO6 (10 Marks)

b. Write a verilog code for SIPO register and PISO register along with circuit diagram and truth table. CO5 (10 Marks)

c. Write a verilog code for PIPO register and SISO register along with circuit diagram and truth table. CO6 (10 Marks)

(10X 2 = 20 Marks)

a. Explain difference between sequential circuits and digital circuits along with examples Explain difference between Latch and flipflop . CO6 (10 marks)

b. Explain and write a verilog code for decade counter. CO5 (10 marks)

c. Write a verilog code for seven segment LED display. Write a short note on test benche CO6 (10 Marks)

Roll No. | | | | | | | |

Paper Code: MGI 102

M.Tech (Geo-Informatics)
Mid Semester Examination 2019
I *Semester*
Fundamentals of Geographic Information Systems

Time: 1:30 Hours

MM: 50

Note:
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks for each main question are ten.

Q1 (2X5=10 Marks)

a "Many definitions have been proposed by different scientists and researchers as per their use" justify the statement by defining GIS as per Good Child and Burrough et.al.?

b Define Point, Line and Mean resolution element? If mean resolution element is 400 consists of 30 states in a country, then what is the area covered?

c "GIS finds its application in all those area where professionals are involved" justify the statement by giving suitable application of GIS in different field?

Q2 (2X5=10 Marks)

a Explain in detail the Essential Components of GIS?

b How GIS applicable for Civil Engineers?

c Define in brief Scanner and Digitizers with their types and method?

Q3 (2X5=10 Marks)

a What are GIS data types? Explain in brief?

b "The data representation is in different kinds of variables, also known as scales that can be stored in a GIS "justify the statement by explain all the variables of Data Representation?

c "The data for GIS collected from different sources" discuss all the sources of data collection?

Q4 (2X5=10 Marks)

a "By and large, the most common application of GIS is the effective management of natural resources and planning of regions at different levels. For such an activity a variety of data se are required" explain in brief these data sets broadly grouped?

b Write a short note on Data Compression, Rastorisation and Vectorisation?

c Define:
1. Data Quality
2. Geo-Referencing

Q5 (2X5=10 Marks)

a Explain in brief with neat diagram conceptual models of spatial information?

b Give the advantages and disadvantages of vector metod?

c Define:
1. Chain Code
2. Quard Tree Tessellation

M.TECH

Roll No. | | | | | | | |

Paper Code: MCS 102

(M.Tech. (CSE)/Ph.D.)

Mid Semester Examination 2019

1st *Semester*

Mobile Computing

Time: 1:30 Hours

MM: 50

Note : (i) This question paper contains two sections.
(ii) Section 'A' is compulsory
(iii) Answer any two questions from Section 'B'
(iv) Each question contains three parts.

Section A (Compulsory)

Q1. (20 Marks)

a. What is wireless technology? Write down growing activity and critical issues related to technology. (7)

b. What are two basic system architecture of IEEE 802.11? How it allow building of Ad-Hoc networks? (7)

c. What are challenges in data management and also describe data replication for mobile computer? (6)

Section B

Q2. (5 X 3 = 15 Marks)

a. What are relative advantage and disadvantage of basic CSMA/CA and CSMA/CA with RTS/CTS protocols? What makes you select one over the others? (5)

b. In a cell with 100 MSs, on an average, if 30 requests are generated during an hour , with average holding time T (360 s), calculate the offered traffic load and also calculate the blocking probability, delayed arriving call probability for a given number of channels (S) in the cell. What would be the efficiency of the system? (5)

c. What is multihopping? Explain multi cluster architecture and problem related to this architecture. (5)

Q3. (5 X 3 = 15 Marks)

a. Looking at the HLR/VLR database approach used in GSM-how does this architecture limit the scalability in terms of users, especially moving users? (5)

b. What are issues in data management? Describe Rumor data replication for mobile computer? (5)

c. How you can compensate for the impact of the Doppler effect in a cellular system? Explain. (5)

Q4. (5 X 3 = 15 Marks)

a. Compare and discuss about reconciliation based data replication and peer to peer based data replication for mobile computer? (5)

b. Explain communication mechanisms in Coda file system? What is the side effect in Coda? (5)

c. Explain the architecture & approach over which Rover toolkit provides mobile Communication support. (5)

Roll No. | | | | | | | |

Paper Code: MSE 102

(*M.Tech*) **Odd Mid Semester Examination 2019**

I *Semester*

Paper Name FEM in Structural Engineering

Time: 1:30 Hours

MM: 50

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks for each main question are ten.

Q1.)

(2X5=10 Marks)

a. Find the displacement in the two-bar assembly loaded as shown below

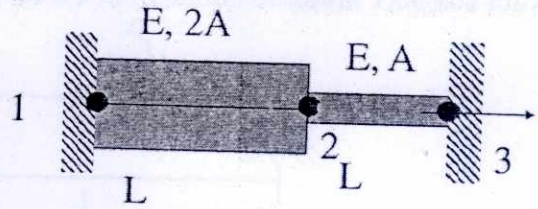


Fig. 1

b. Compute the stresses in the two-bar assembly in fig. 1.

c. Compute the stiffness matrix for the truss below fig.2

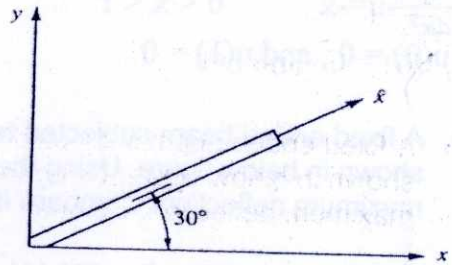
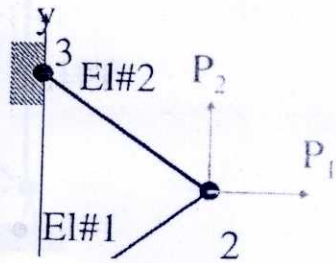


Fig. 2

Q2.

(2X5=10 Marks)

a. The length of bars 12 and 23 are equal (L) E: Young's modulus A: Cross sectional area of each bar Solve for (1) d2x and d2y



b. Calculate stresses in each bar of fig. 3 assuming the same data as given in (a) part.

c. Justify how galerkin method is the most accurate method among all the variational method.

Q3. (2X5=10 Marks)

a Explain the different weighted residual methods. Which one of them is the accurate method for solving differential equation.

b. Calculate the displacements at node 2 and node 3 of the spring given below

c.

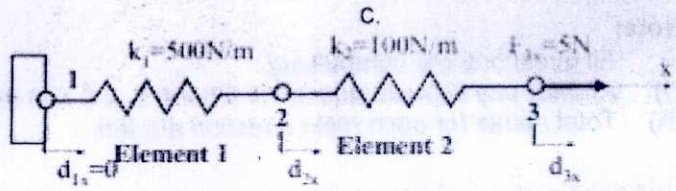


Fig. 4

c . Using any FEM analysis calculate the deflection at point P and compare the solutio with exact solution. Assume equation of curve is y=ax(L-x)

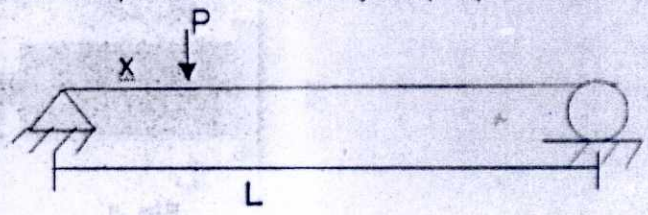


Fig. 5

Q4. (2X5=10 Marks)

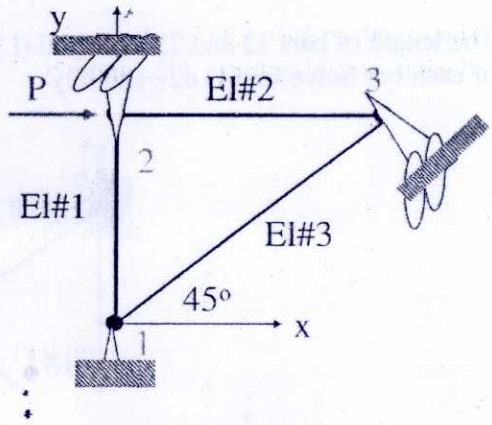
a. Consider the following sample problem Solve by any weighted residual method

$$\frac{-d^2u}{dx^2}-u=-x \qquad 0<x<1$$

$u(0) = 0$, and $u(1) = 0$

b. A fixed ended beam subjected to uniformly distributed load of intensity p is shown in below figure. Using the Rayleigh-Ritz procedure, compute the maximum deflection. Compare the results with analytical solution.

c. For the plane truss P=1000 kN, L=length of elements 1 and 2 = 1m E=21 GPa , A = $6\times10^{-4}m^2$ for elements 1 and 2, A=2×10^{-4} m^2 for element 3. Determine the unknown displacements.



Q5 (2X5=10 Marks)

a. What are the merits and de merits of finite element method.

b. For the bar shown in figure determine the nodal displacements at node 2 and node 3.

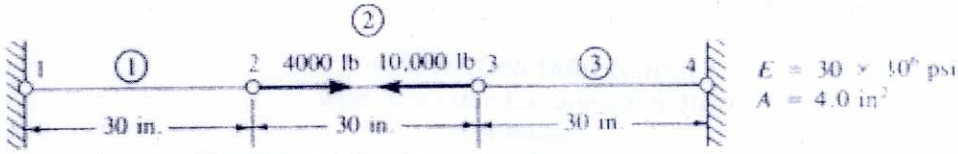


Fig. 7

c. Calculate the stresses and reactions shown in fig.7.

M.TECH

Roll No. | | | | | | | |

Paper Code: MSE 101

M.Tech (Structures)

Odd Mid Semester Examination 2018

I Semester

Paper Name :Theory of Elasticity and Plasticity

Time: 1:30 Hours

MM: 50

Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks for each main question are ten.

Q1. **(5 X 2 =10 Marks)**

a. Describe the ideology behind various assumptions made in the theory of elasticity? (5)

b.The state of stress at a point with respect to 'xyz' Cartesian coordinate system were given as $\begin{bmatrix} 15 & 10 & -10 \\ 10 & 10 & 0 \\ -10 & 0 & 40 \end{bmatrix}$ Determine the magnitude of Normal and shear stress on a surface intersecting the point parallel to the plane given by 5x-2y+3z=0? (5)

c. Derive the Stress Transformation equation? (5)

Section B

Q2. **(5X2= 10 Marks)**

a. Write the steps for Mohr's Circle construction and how it is useful for Principal strain determination? (5)

b. Derive the equilibrium equations for plane strain condition? (5)

c. The state of stress at a point is given by $\begin{bmatrix} 9 & 6 & 5 \\ 6 & 5 & 2 \\ 5 & 2 & 4 \end{bmatrix}$ determine the principal stresses and principal directions? (5)

Q3. **(5X2= 10 Marks)**

a.Explain your understanding on Octahedral stresses? (5)

b.Principal Stresses for Plane Strain condition (5)

c.Hooks Law and Linear relations between the components of stress and the components of strain? (5)

Q4. **(5 X 2 = 10 Marks)**

a. Write the equations for principal stresses and Stress invariants? (5)

b. If the displacement field in a body is specified by $u = (2x^2 + 3)10^3$, $v = (3y^2z)10^3$ and $w = (2x + 3z)10^3$, determine the strain components at a point whose coordinates are (3,2,3)? (5)

c. Derive the equation of compatibility for strains? (5)

Q5. **(5 X 2 = 10 Marks)**

a. Explain the airy stress fuction? (5)

b. Write the stress components on a rectangular parallelepiped in Cartesian coordinate system (5)

c. Derive the equation of compatibility for stresses? (5)

Roll No. | | | | | | | |

Paper Code: MCS-101

(*M.Tech-CSE*) Odd Mid Semester Examination 2019

I- *Semester*

Paper Name: Advance Algorithms

Time: 1:30 Hours

MM: 50

Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks for each main question are ten.

Q1.) (2X5=10 Marks)

a. Give the Prims algorithm to find minimal spanning tree in given connected weighted graph

b:. Compare the order of growth of functions, 5nlogn, n^2 and 3^n

c: Prove that if $t_1(n) \in O(g_1(n))$ and $t_2(n) \in O(g_2(n))$ then show that $(t_1(n).t_2(n)) \in O\{(g_1(n).(g_2(n)\}$

Q2. (2X5=10 Marks)

a. Discuss the time complexity of Insertion sort algorithm in its worst case

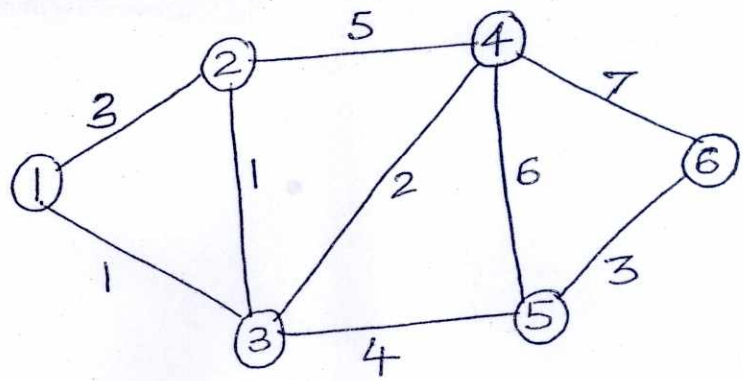
b. Define fractional knapsack problem. Solve the given instance using the greedy approach; the capacity of the knapsack is 35.

Item	Weight	Value in Rs
1	30	300
2	20	220
3	10	160

c:. Give big O estimation of function $9n^3+6n + n^2\log n$.

Q3. (2X5=10 Marks)

a:. Use the Dijkastra algorithm to get shortest path in a given connected weighted graph, Consider node 1 as source.



b: Solve the recurrence relation T (n) =3T (n/4) + cn with help of recursion tree.

c:. What is time complexity and space complexity of an algorithm? Explain big O and big Ω notation with example.

.

Q4. (2X5=10 Marks)

a: Discuss the Merge Sort algorithm, Set recurrence relation for Merge sort and compute the time complexity.

b:. Give the Greedy algorithm to solve the travelling Salesman problem. Explain with help of an example.

c:. Give the Rabin Karp algorithm of string matching.

Q5 (2X5=10 Marks)

a. Explain the followings.

i): Reducibility ii) NP Problems iii) Satisfiability problem iv) Max Clique Problem.

b. Show that amortized cost of a sequence of n operation on a stack is O(n)

c. Discuss the Depth First Search technique.

Paper Code: MGI 101

M. Tech.

Mid Semester Examination 2019

1st Semester

Fundamental of Remote Sensing

Time: 1:30 Hours MM: 50

Note: This question paper contains two sections. Section *A* is compulsory and answer any two from Section *B*.

Section A (Compulsory)

1. (a) Differentiate between the active and passive remote sensing. What are the advantages of active remote sensing over passive remote sensing? (7)

   (b) What are different types of scattering that affect remote sensing images? (7)

   (c) Write the advantage and disadvantage of remote sensing. (6)

Section B (Answer any two)

2. (a) Explain ideal remote sensing and its shortcomings. (5)

   (b) Write the wavelength range of optical, infrared and thermal infrared useful in remote sensing. (5)

   (c) Write at least ten application of remote sensing. (5)

3. (a) Provide the details (description of spectral band and all kind of resolution) of any three remote sensing satellites. (5)

   (b) What do you understand by the multi concept in remote sensing? (5)

   (c) What are the obvious sources of electromagnetic energy in remote sensing? (5)

4. (a) Explain the different kind of resolution in remote sensing. (5)

   (b) Discuss the radiation from a blackbody and its dependence of temperature. (5)

   (c) What do you understand by spectral reflectance curve? Draw the spectral reflectance cure for water. (5)

M.TECH

Roll No. | | | | | | | |

Paper Code: MSE 101

M.Tech (Structures)

Odd Mid Semester Examination 2018

I Semester

Paper Name :Theory of Elasticity and Plasticity

Time: 1:30 Hours

MM: 50

Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks for each main question are ten.

Q1. **(5 X 2 =10 Marks)**

a. Describe the ideology behind various assumptions made in the theory of elasticity? (5)

b.The state of stress at a point with respect to 'xyz' Cartesian coordinate system were given as $\begin{bmatrix} 15 & 10 & -10 \\ 10 & 10 & 0 \\ -10 & 0 & 40 \end{bmatrix}$ Determine the magnitude of Normal and shear stress on a surface intersecting the point parallel to the plane given by 5x-2y+3z=0? (5)

c. Derive the Stress Transformation equation? (5)

Section B

Q2. **(5X2= 10 Marks)**

a. Write the steps for Mohr's Circle construction and how it is useful for Principal strain determination? (5)

b. Derive the equilibrium equations for plane strain condition? (5)

c. The state of stress at a point is given by $\begin{bmatrix} 9 & 6 & 5 \\ 6 & 5 & 2 \\ 5 & 2 & 4 \end{bmatrix}$ determine the principal stresses and principal directions? (5)

Q3. **(5X2= 10 Marks)**

a.Explain your understanding on Octahedral stresses? (5)

b.Principal Stresses for Plane Strain condition (5)

c.Hooks Law and Linear relations between the components of stress and the components of strain? (5)

Q4. **(5 X 2 = 10 Marks)**

a. Write the equations for principal stresses and Stress invariants? (5)

b. If the displacement field in a body is specified by $u = (2x^2 + 3)10^3$, $v = (3y^2z)10^3$ and $w = (2x + 3z)10^3$, determine the strain components at a point whose coordinates are (3,2,3)? (5)

c. Derive the equation of compatibility for strains? (5)

Q5. **(5 X 2 = 10 Marks)**

a. Explain the airy stress fuction? (5)

b. Write the stress components on a rectangular parallelepiped in Cartesian coordinate system (5)

c. Derive the equation of compatibility for stresses? (5)

Roll No. | | | | | | | |

Paper Code: MCS-101

(*M.Tech-CSE*) Odd Mid Semester Examination 2019

I- *Semester*

Paper Name: Advance Algorithms

Time: 1:30 Hours

MM: 50

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks for each main question are ten.

Q1.) (2X5=10 Marks)

a. Give the Prims algorithm to find minimal spanning tree in given connected weighted graph

b:. Compare the order of growth of functions, 5nlogn, n^2 and 3^n

c: Prove that if $t_1(n) \in O(g_1(n))$ and $t_2(n) \in O(g_2(n))$ then show that $(t_1(n).t_2(n)) \in O\{(g_1(n).(g_2(n)\}$

Q2. (2X5=10 Marks)

a. Discuss the time complexity of Insertion sort algorithm in its worst case

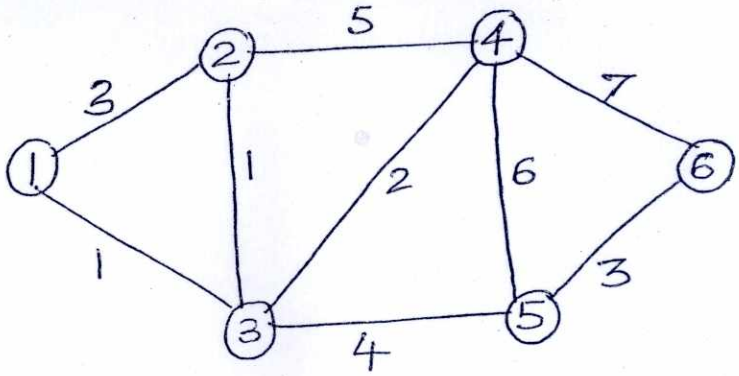
b. Define fractional knapsack problem. Solve the given instance using the greedy approach; the capacity of the knapsack is 35.

Item	Weight	Value in Rs
1	30	300
2	20	220
3	10	160

c:. Give big O estimation of function $9n^3+6n + n^2\log n$.

Q3. (2X5=10 Marks)

a:. Use the Dijkastra algorithm to get shortest path in a given connected weighted graph, Consider node 1 as source.



b: Solve the recurrence relation T (n) =3T (n/4) + cn with help of recursion tree.

c:. What is time complexity and space complexity of an algorithm? Explain big O and big Ω notation with example.

(2X5=10 Marks)

Q4.

a: Discuss the Merge Sort algorithm, Set recurrence relation for Merge sort and compute the time complexity.

b:. Give the Greedy algorithm to solve the travelling Salesman problem. Explain with help of an example.

c:. Give the Rabin Karp algorithm of string matching.

(2X5=10 Marks)

Q5

a. Explain the followings.

i): Reducibility ii) NP Problems iii) Satisfiability problem iv) Max Clique Problem.

b. Show that amortized cost of a sequence of n operation on a stack is O(n)

c. Discuss the Depth First Search technique.